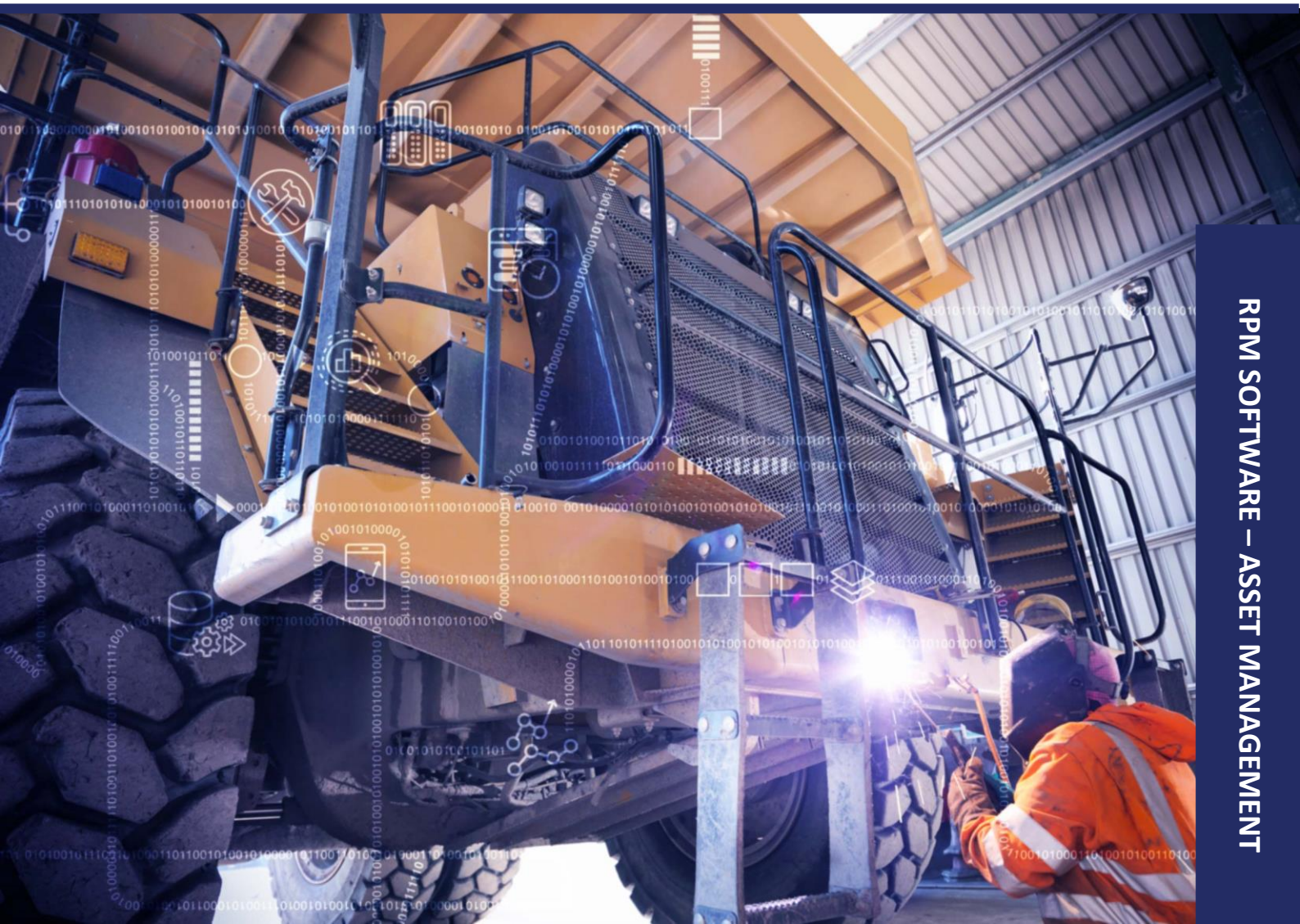
RPMGLOBAL

AMT – Long Term Planning

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task



Important Reminder

An important reminder for the participant about the task



Learning Check

Check your understanding on the workshop and ask any related questions.

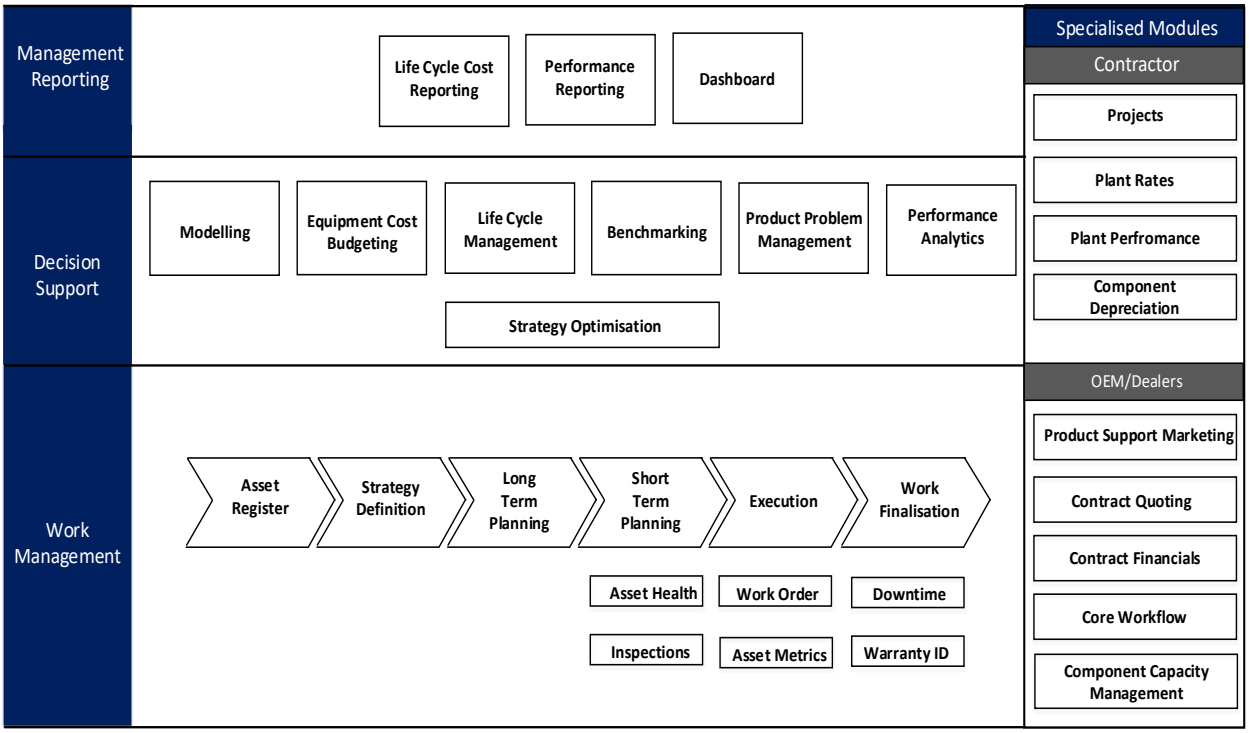
Course Outline

Prerequisites	Getting Around AMT, AMT Key Concepts
Target Audience	Reliability Engineer, Senior Planner, Short Term Maintenance Planner
Nominal Duration	
Course Objectives	The objectives of the course are: • Implement the maintenance strategy • Optimise component lives • Optimise scheduling efficiency • Manage resources • Maintain equipment availability • Coordinate maintenance activities
Learning Outcomes	At the end of the course, participants will have learned to: • Review Actual Usage Readings ○ Use Record Actual Usage function ○ Delete an Invalid Usage reading ○ Add a New Actual Usage reading ○ Use the Quick Watch function ○ Run Reports to Check Actual Usage Readings ○ Run Fleet Statistics report ○ Procedure - Utilisation Analyser Screen

	○ Edit a Usage Profile ○ Run Usage v Profile Snapshot report • Review Overdue Tasks ○ Review overdue tasks from the My Functions hierarchy ○ Review overdue tasks using Quick Watch • Review New Tasks ○ Review New Tasks • Monitor Existing Tasks ○ Monitor Existing Tasks • Set Date for Tasks within Twelve Weeks (3 month look ahead) ○ Set Date for Tasks • Optimise Timing (Group/Align Tasks) ○ Group Tasks • Validate Parts and Components ○ Validate Parts and Components • Validate Labour Resources ○ Validate Labour Resources • Register a Manual Component Changeout ○ Register a Manual Component Changeout • Confirm Component Changeout Schedule ○ Confirm Component Changeout Schedule • Review forecasts • Forecast Availability Report ○ Forecast Availability Report • Cost Forecast Reports ○ Forecast Cost Analysis report ○ Forecast Cost Worksheet report ○ Cost Analysis – “Life Cycle” report • Component Forecast Reports ○ Component Status report ○ Overdue Strategy Tasks report ○ Changeout Analysis - Detailed report • Labour Forecast Reports ○ Labour Forecast - Hours report ○ Workload Optimisation report • Long Term Planning Meeting • Confirm Plan Communicate to Stakeholders
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system that consists of integrated modules as depicted below. The functionality covered by this training manual is identified by the Long-term Planning section in the diagram.



Overview of Long-term Planning

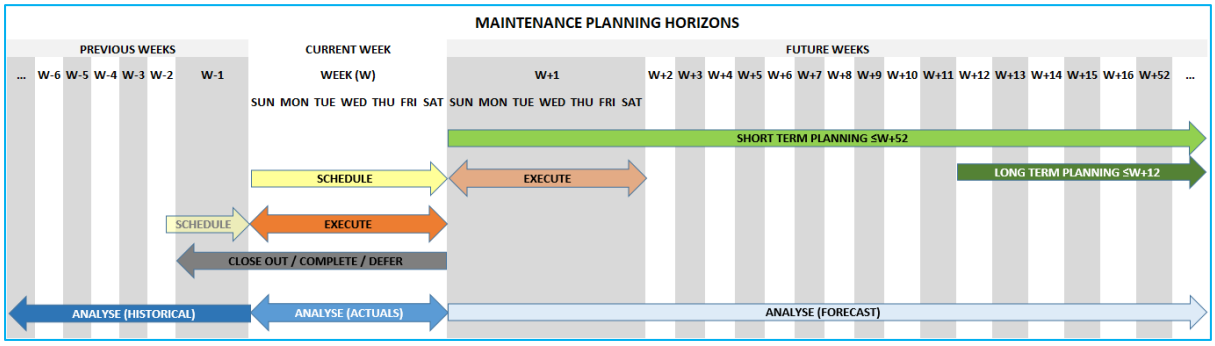
Long-term planning is based on the concept of ensuring that major maintenance (strategic) tasks have minimal impact on your company's production or operations targets. To expand on this concept, it is the formal identification of the major strategy activities and ensuring minimal impact on the production and cost targets by optimising labour resources, parts, materials and tooling through accurate forecasting.

The Long-term Planning module is used to analyse and produce the following information:

- Produce a W+12 forecast for the short-term planning and W+1 scheduling cycles.
- Provide multiple inputs for the annual budgeting and forecasting cycle.
- Availability requirements forecast.
- One year and beyond availability or utilisation projections.
- Manage lead times on long lead time components.
- Identify opportunities to optimise the DLCC.
- Assist in identifying fleet replacement opportunities.

The main focus of the procedures in the following pages has been the first two points - producing a W+12 forecast and providing the inputs for the annual budgeting cycle.

Current best practice is to update and produce the W+12 (12 week / 3 month) forecast weekly as an integral part of the short-term scheduling process; that is, the W+1 maintenance schedule. The maintenance planning horizons are explained in the diagram below.



The plans are reviewed with the operations team as part of the initial draft of the maintenance schedule and, along with the PM services, forms the basis of next week's (W+1) schedule.

Strategy changes that only apply to the next occurrence of a strategy task are called long-term planning adjustments. For example:

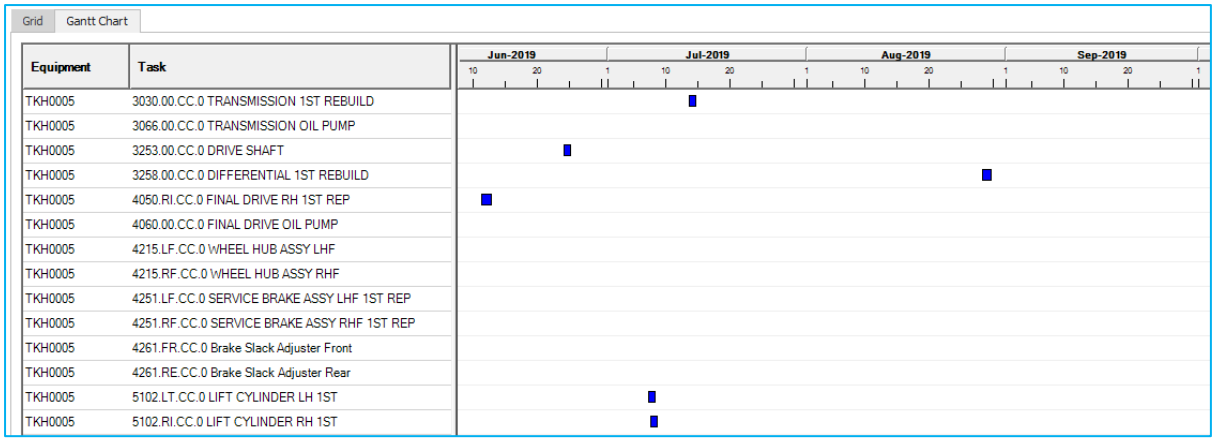
- For the transmission in a loader, based on condition monitoring analysis, the component may need to be replaced three months earlier than expected.
- For the suspension on a truck, there may not be any rotable or service exchange components available when the suspension is due for replacement in three months, therefore, new components will have to be used.

These strategy changes only affect the next occurrence – that is, the strategy for the components that are currently in the machine. Changes to long-term planning only affect the next occurrence; that is, they do not change the parameters of the overall maintenance strategy frequency or interval.

The **Long-term Planning** grid shows all planned strategy tasks for the equipment selected in the filters (non-planned strategy tasks are excluded).

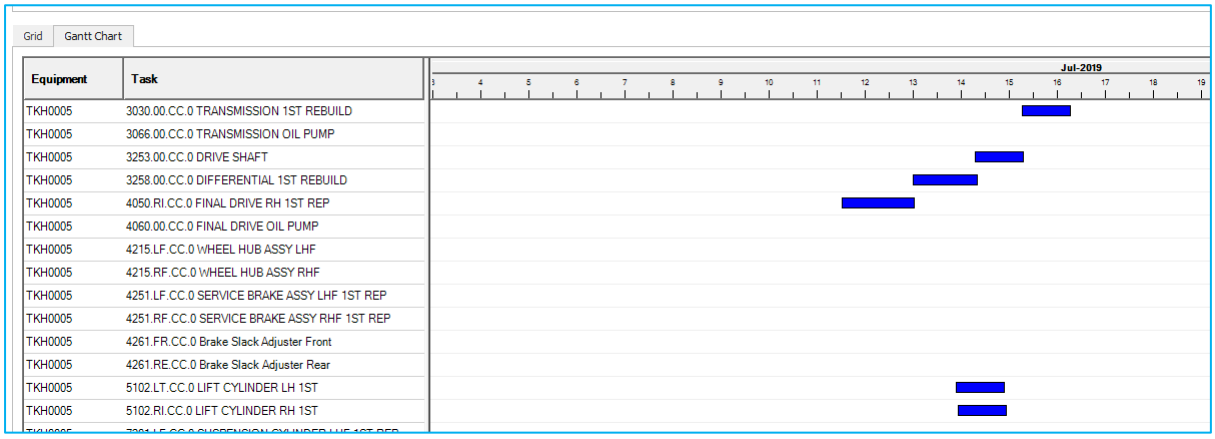
Part of the long-term planning process is aligning or staging these strategy tasks at a high level. For example, a truck may have the following components due for replacement within three months of each other:

- Main drive shaft
- Transmission
- Right final drive
- Both lift (hoist) cylinders
- Differential



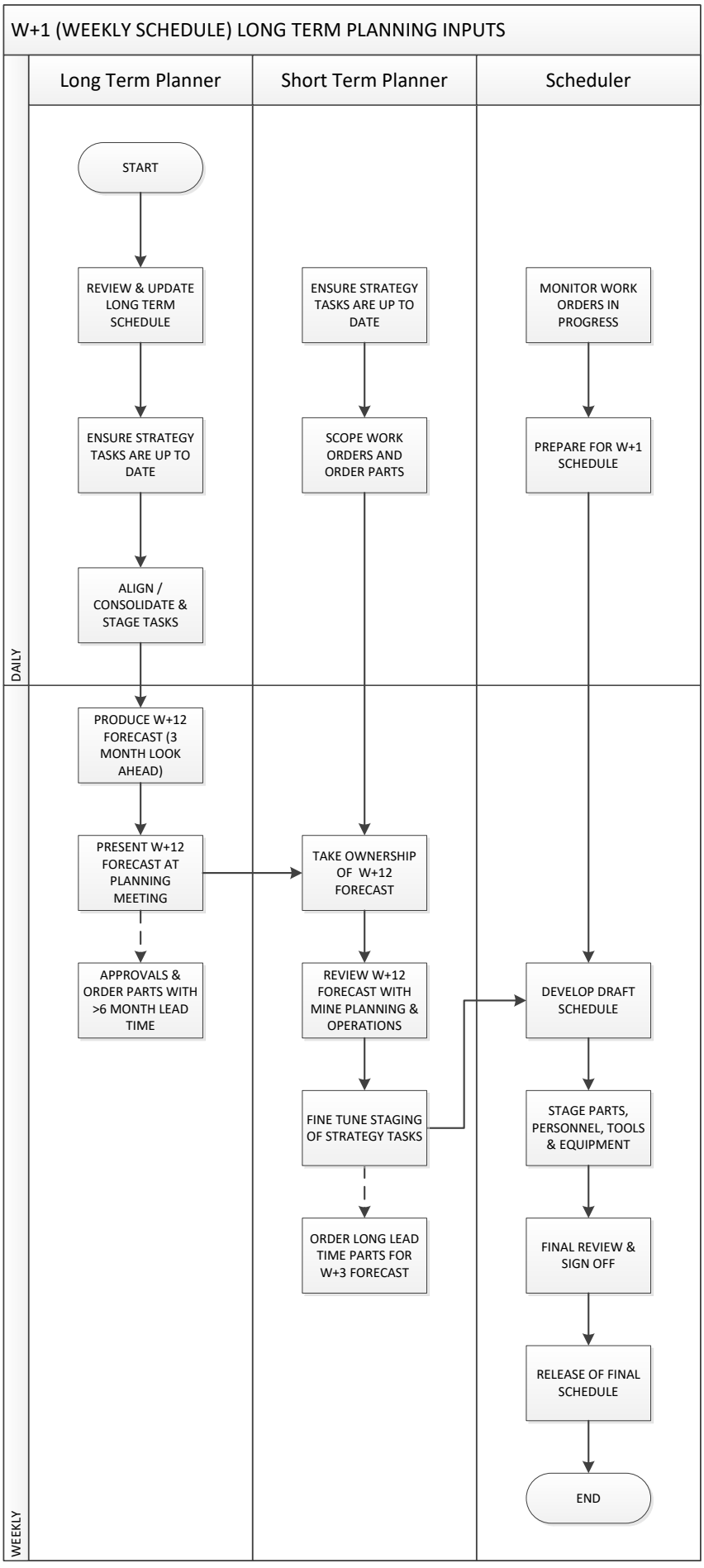
Instead of executing each strategy task individually over a longer period, the long-term planner tries to align and stage these tasks to increase uptime by consolidating all of these tasks into one major outage.

There are many other factors to consider here. For example, operations may be at a particularly critical phase of the mine plan at this point, and a longer outage is not feasible for compliance with the mine plan, despite providing better availability overall. The long-term planner may have to stage the outages into two or more shorter events to ensure compliance, even though it reduces the maintenance efficiency in the process.



After the long-term planner has done some high-level staging of the strategy tasks to consolidate the complimentary tasks, the W+12 forecast is given to the short-term planner, who fine-tunes the staging of the components, parts, and services for anything that falls within the W+12 (12 week) forecast.

When these tasks have been planned and the components and parts ordered, the scheduler takes control of the strategy tasks and includes them in the W+1 (weekly maintenance schedule). The long-term planning process is illustrated in the following process flows.


W+1 (WEEKLY SCHEDULE) LONG TERM PLANNING INPUTS
Long Term Planner
Short Term Planner
Scheduler
DAILY
WEEKLY
START
REVIEW & UPDATE LONG TERM SCHEDULE
ENSURE STRATEGY TASKS ARE UP TO DATE
ALIGN / CONSOLIDATE & STAGE TASKS
ENSURE STRATEGY TASKS ARE UP TO DATE
SCOPE WORK ORDERS AND ORDER PARTS
MONITOR WORK ORDERS IN PROGRESS
PREPARE FOR W+1 SCHEDULE
PRODUCE W+12 FORECAST (3 MONTH LOOK AHEAD)
PRESENT W+12 FORECAST AT PLANNING MEETING
APPROVALS & ORDER PARTS WITH >6 MONTH LEAD TIME
TAKE OWNERSHIP OF W+12 FORECAST
REVIEW W+12 FORECAST WITH MINE PLANNING & OPERATIONS
FINE TUNE STAGING OF STRATEGY TASKS
ORDER LONG LEAD TIME PARTS FOR W+3 FORECAST
DEVELOP DRAFT SCHEDULE
STAGE PARTS, PERSONNEL, TOOLS & EQUIPMENT
FINAL REVIEW & SIGN OFF
RELEASE OF FINAL SCHEDULE
END

Login and Navigation

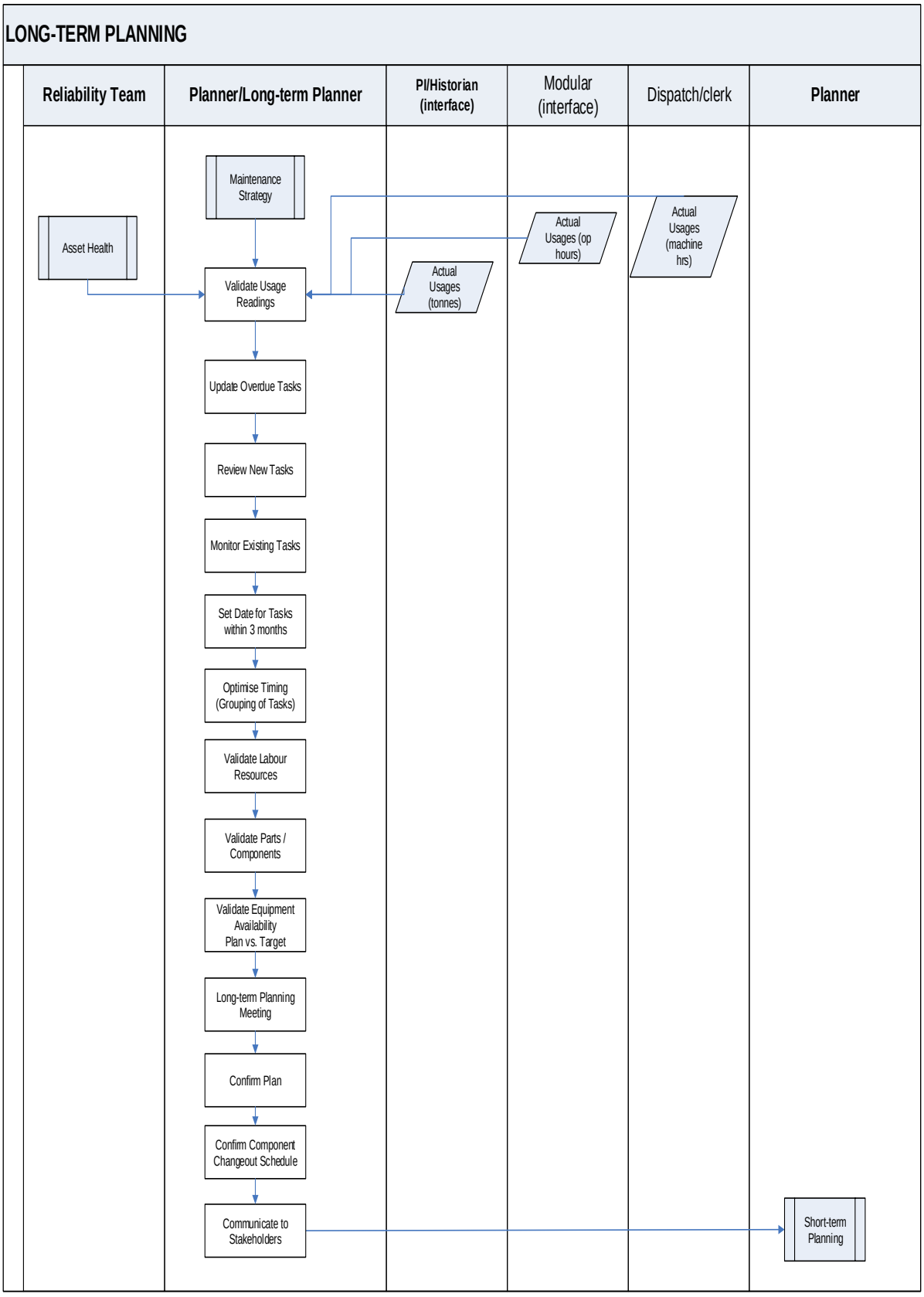
For information on AMT login and navigation, refer to the AMT Training Manual - Getting Around AMT.

Glossary

For information on AMT terminology, refer to the AMT Glossary, which is available as a handbook as well as in the AMT online help.

Processes

Process Diagram



Process Owner

The owner of the process is the Senior (or Long-term) Planner.

Process Inputs

The following inputs are required for this process:

- Strategy definition
- Up-to-date utilisation
- Standard jobs
- Site labour levels
- Availability targets
- Condition monitoring data

Process Outputs

The following outputs can be expected from this process:

- Component changeout schedule for the next year
- Confirmed component changeout schedule for the next 12 weeks
- Accurate parts and labour forecasts
- Availability forecast

Training Workshops

1. Review Actual Usage Readings

Overview

This step is only applicable for assets based on usage.

You can use various reports to review that the actual usage readings exist in AMT. Actual usage data informs AMT where an asset is in its life cycle, directly impacting forecasting.

There are two ways to do this in AMT:

- Use the **Record Actual Usages** function.
- Use the **Quick Watch** function.

Procedure – Use Record Actual Usage function

Perform the following steps using the **Record Actual Usages function.**

1. Open the **Record Actual Usages** screen.



*Alternatively, from the **My Functions** hierarchy, select **Daily Operations > Usages & Measurements > Record Actual Usages.***

2. On the **Actual Usages** screen, filter for the **Site**, **Fleet**, **Model**, **Equipment** and **UOM**.
3. Set the date range to the last seven days and apply the filter.
4. Sort the grid by the **Date Time** column to see the most recent reading (if manual readings are updated daily).
5. Click **Apply Filter**.
6. On the **Equipment Actual Usages** screen, validate the following columns:
    - **Equipment**, its **UOM** and the **Date Time** of the last reading.
    - **Reading**, **Offset** and **Cumulative Usage:** The values in the **Reading** and **Cumulative Usage** columns are usually the same. They could be different during certain scenarios. For example, if the meter that is recording usage, let's say hours, is replaced, its readings could go back to zero. The equipment would have hours of usage. In this case, the **Offset** is mentioned in the corresponding column.
    - **Exceeds (1) Projection:** All usage-based equipment will have a usage assumption. If the actual reading exceeds these assumptions, the equipment is marked accordingly in this column. This is just a warning and does not stop any function in AMT.
    - **Invalid (2):** AMT flags a reading as "invalid" in two scenarios:
        - When a reading is less than the previous reading. Usage readings must always increase, as they are cumulative.
        - When the new reading is higher than the threshold value set in AMT. AMT has been configured with a threshold of 500% of the expected usage.
7. Click **Save** before exiting the **Actual Usages** dialog.

You can right-click on the grid and copy the data into a spreadsheet program. In the spreadsheet, you can complete New Date Time, New Reading and/or New Offset value and then paste the full grid back into AMT.

Procedure – Delete an Invalid Usage reading

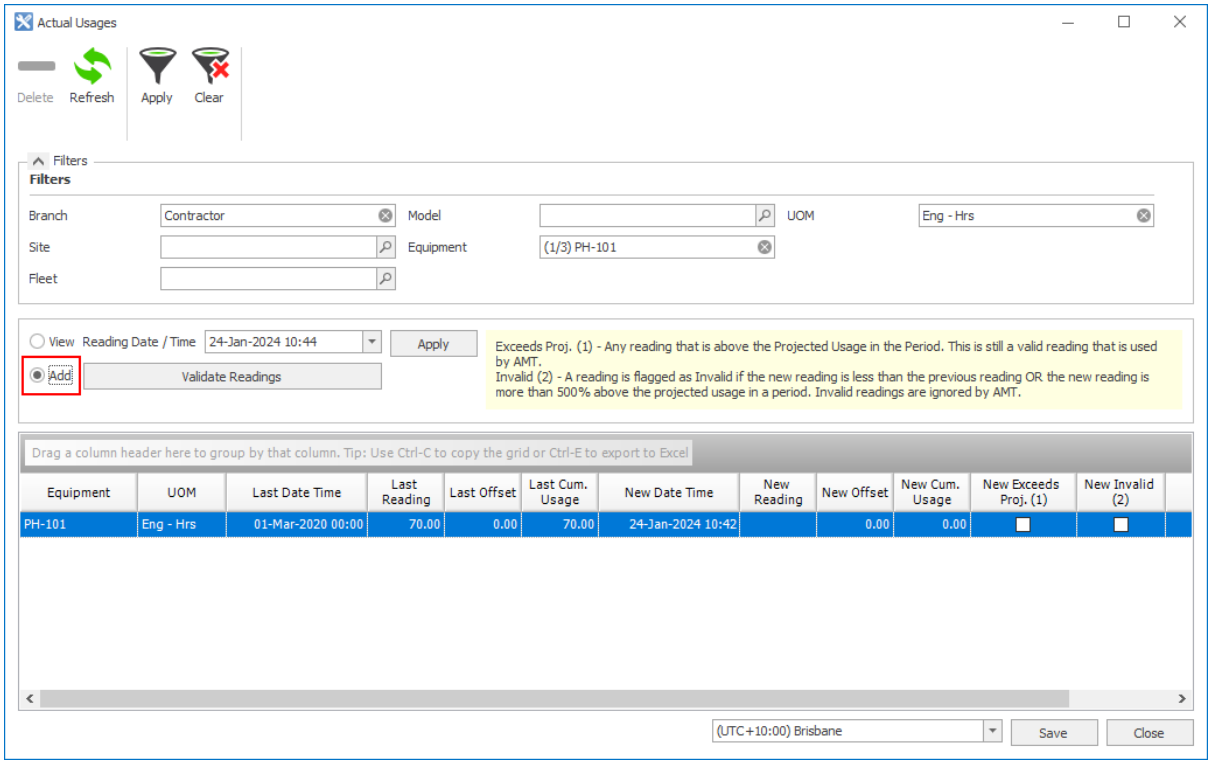
Perform the following steps to delete an invalid usage reading or add a new actual usage reading:

1. Select the row and click **Delete**.
2. Click **Yes** to confirm the deletion in the notification box.
3. Click **Save**.

Procedure – Add a New Actual Usage reading

Perform the following steps to add a new actual usage reading.

1. Filter for the appropriate **Equipment** and **UOM**.
2. Select **Add**.



3. Set the appropriate new date and time in the **View Reading Date / Time** field and click **Apply**.
4. Enter the **New Reading** value in the corresponding column.
5. Click **Validate Readings**.

6. If the reading is not invalid, click **Save**.

Procedure – Use the Quick Watch function

Perform the following steps using the **Quick Watch** function.

1. On the AMT Home page, click the arrow next to **Quick Watch.**
2. From the Quick Watch options, select **No Recent Usage Readings.**
3. Select the desired filters and click **Apply Filter**.



*You can only click **Apply Filter** in the Quick Watch area after selecting a Quick Watch option.*

4. A list of assets that have not received actual usages displays.

Procedure – Run Reports to Check Actual Usage Readings

You can run the following reports in AMT to ensure the actual usages readings are correct: Fleet Statistics report, the Utilisation Analyser screen and the Usage v Profile Snapshot report.

Procedure – Run Fleet Statistics report

This report shows the latest actual usage readings for an equipment.

Perform the following steps to run the Fleet Statistics report:

1. Search **Fleet Statistics** using the **Search** field and double-click **Fleet Statistics**.



*Alternately, from the **My Functions** hierarchy, select **Daily Operations** > **Usages & Measurements** > Fleet Statistics.*

2. Select the relevant **Fleet** to run the report at the fleet level.
3. Set the relevant **UOM** (such as **Eng Hrs** or **Op Hrs**).
4. Click **Preview**.
5. The **Fleet Statistics** report displays.

Procedure - Utilisation Analyser Screen

This screen allows you to compare actual utilisation with forecast utilisation and quickly make adjustments to forecast utilisation.

Perform the following steps to access the **Utilisation Analyser** screen:

1. Search **Utilisation Analyser** using the **Search** field and double-click **Utilisation Analyser**.



*Alternatively, from the **My Functions** hierarchy, select **Daily Operations** > **Usages & Measurements** > **Utilisation Analyser.***

2. Click the **Advanced Utilisation** tab for assets configured in advanced utilisation mode or click the **Simple Utilisation** tab for assets configured in simple utilisation mode.
3. Select the relevant **Fleet** and **Days to Analyse Actuals** (that is, on how many days you want to review).
4. Click **Apply Filters**.
5. On the **Utilisation Analyser** screen, validate the columns:
    - **Daily Average Actual Usage** comes from record actual usages.
    - **Annual Average Actual Usage** refers to how many expected hours the asset will do based on actuals.
    - **Current Annual Forecast Usage** is the hours from the mine plan in your usage profile for the year.

Procedure – Edit a Usage Profile

Perform these steps to edit a usage profile:

1. Search **Usage Profile** using the **Search** field.



*Alternatively, from the My functions hierarchy, select **Setup & Admin** > **Usage Profiles** > **Usage Profile.***

2. Select **Equipment**, Set **Projection** as **Current** and click **OK**.
3. The **Edit Usage Profile** dialog displays. You can overwrite the existing step usage rate or enter a new usage step with a new rate.
4. Click **Add** in the **Usage Steps** section.
5. On the **Add Usage Step** dialog, enter the relevant **Step Start Date** and click **Save**.
6. Repeat step 4 – 5 as needed to add more usage steps.
7. Click **Save**.

Procedure – Run Usage v Profile Snapshot report

This report compares the usage of the equipment based on the UOM with the target. This could highlight issues with the application or the client machine usage.

Perform the following steps to access the Usage v Profile Snapshot report:

1. Search **Usage v Profile Snapshot** using the **Search** field and double-click **Usage v Profile Snapshot**.



*Alternatively, from the My functions hierarchy, select **Reporting & Analyses** > **Equipment Performance** > **Utilisation** > **Usage v Profile Snapshot.***

2. On the **Usage v Profile Snapshot - Report Parameters dialog**, perform the following actions:
    - Select the **Last 12 + Next 6 Months** checkbox.
    - Select the relevant **Fleet**.
    - Set **UOM** as the relevant UOM (such as **Eng Hrs** or **Op Hrs**).
    - Click **Preview**.
3. The **Usage v Profile Snapshot** report displays.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

2. Review Overdue Tasks

Overview

Long-term planning is done for large tasks such as component changeouts or equipment shutdowns.

As the second step, we check for and update any overdue strategy tasks. Overdue tasks are tasks that should have been done by now but have not been completed. For an overdue task, AMT assumes that the task is due on the present day.

The question is: why is the task overdue? Validate why the task has not been rescheduled to when the site plans to perform the work. Residual value in component (RVIC) and projected cost variance (PCV) are analysed in this step.

There are two ways to view overdue tasks:

- From the My Functions hierarchy
- Using Quick Watch

Procedure – Review overdue tasks from the My Functions hierarchy

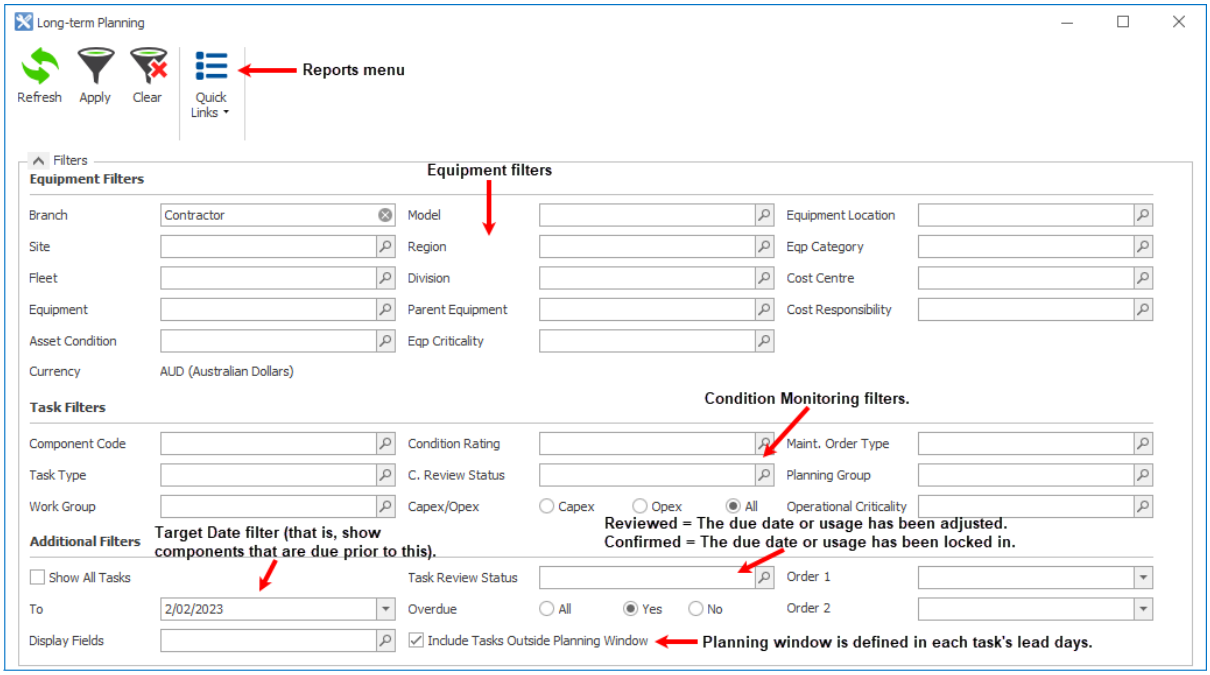
Perform the following steps to review overdue tasks from the **My Functions** hierarchy:

1. Open the **Long-term planning** screen.



Alternatively, from the My Functions hierarchy, select ***Long-term Planning > Long-term Planning.***

2. On the **Long-term Planning** dialog, perform the following actions:
    - Select **Branch**, **Site**, **Fleet**, **Model** for the equipment.
    - Use **Task Filters** to filter by **Component Code** and **Task Type**.
    - Select dates range.
    - In the **Display Fields** field, you can select **Life used %**, **Days Remaining**, **Cost**, **Risk Rating (Calculated)** or **Risk Override**.
    - Set **Overdue = Yes**.
3. Click **Apply Filters**.



4. Minimise the filters by clicking the upwards chevron beside **Filters**.
5. Tasks appear in the **Planning** grid based on the filters. Overdue tasks are identified by a red exclamation mark **!** in the **Overdue** column.

   Additional information that may help you to understand the current status of the tasks are the following columns:

   - **Frequency:** Maintenance task frequency.
   - **Life to Date:** For overdue tasks, this will be greater than the frequency.
   - **Life used %:** For overdue tasks, this will be greater than 100%.
   - **Days Remaining:** For overdue tasks, this will be negative and thus reflects the number of days the task is overdue.
   - **Cost:** Cost of the task, which will help you to prioritise which overdue tasks to investigate.
   - **Risk Calculated:** Auto-calculated risk based on the combination of Equipment Criticality-Strategy Task Operational Criticality-Asset Condition.
   - **Risk Override:** Manually entered risk rating.
   - **System Calc:** If the checkbox is selected, AMT has calculated the strategy task's next occurrence date or usage.

   Having prioritised the items, the planner should determine why the tasks are overdue in order to understand the context.

   Validate the following entries in the list:

   - **Life to Date:** The current hours minus last completed hours.
   - **Strategy Date/Usage:** The date when the task is due next.
     - For a date-based task, the relevant date is updated; for a usage-based task, the relevant usage is updated.
6. To reschedule an overdue task, adjust the **Strategy Date** or the **Strategy Usage** columns by clicking in the appropriate cell on the **Long-term Planning** grid and selecting or typing a new value.

7. After changes have been made, review status by going to the **Review Status** column and changing the status to **Reviewed** or **Confirmed**.



Reviewed *= You've adjusted the* ***Due*** *Date/Usage. It's updated but is pending final information before confirming.*

Confirmed *= You've locked in the* ***Due*** *Date/Usage. The Target Date/Usage is accurate and short-term planners should work to this target.*

8. Below the grid is the **Changeout Reason** field. This a free-text field where you can enter notes relevant to the selected task; it is not exclusively restricted to a component replacement.
9. After you've finished your changes, click **Save** to save and close the **Long-term Planning** grid. If you haven't made any changes, then click either **Save** or **Close** to exit the **Long-term Planning** screen.

Procedure – Review overdue tasks using Quick Watch

You can also use the **Quick Watch** function to view overdue tasks.

Perform the following steps to review overdue tasks using Quick Watch:

1. Click the **Overdue Component Changeouts** option in the **Quick Watch** menu.

   AMT will populate the corresponding list.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

3. Review New Tasks

Overview

It is important for the personnel responsible for reviewing the long-term plan to prioritise. A single equipment can be made up of hundreds of tasks (components replacements, services, and so on), which quickly multiplies when assessing the long-term plan of a fleet.

In long-term planning, switching off the **Include Tasks outside Planning Window** filter allows the planner to focus on the tasks that have entered the planning window because they are the highest priority.

Procedure – Review New Tasks

Perform the following steps to review new tasks in the **Long-term Planning** screen:

1. The planning window's timeframe is defined for each strategy task in the **Planning Lead Time** field of the **Strategy Task** window. The timeframe is either a system default value or can be defined during task creation.
2. The lead time set by the user (typically set by the corporate equipment management group) usually spans the time taken to order the part, component, or external service, plus a buffer to evaluate the condition of the component or allow for potential delays in coordination. This lead time will be site-specific.
    - Back on the Long-term planning screen, filter for the required **Equipment**.
    - Set the filters to see the strategy tasks requiring immediate attention.
    - **Include Tasks Outside Planning Window = No**.
3. Click **Apply Filters**.



Excluding tasks outside the planning window means that strategy tasks that have not yet reached their respective lead days are not shown.

4. To set the next occurrence based on usage (such as engine hours):
    i. Highlight the task in the grid.
    ii. Click **Strategy Usage**.
    iii. Enter the target usage (that is, the equipment's usage when the strategy task will be performed).

    Alternatively, to set the next occurrence based on date:
    i. Highlight the task in the grid.
    ii. Click **Strategy Date**.
    iii. Enter the target date when the strategy task will be performed.

Change the **Review Status** to be either:

- **Reviewed** –typically means that a target changeout point has been entered, but for example, the component's health is still being assessed.
- **Confirmed** – the target date/usage is accurate, and planners should work to this target.

5. You can run a range of reports to show the adjustments made in long-term planning. Access the reports through the menu hierarchies on the main screen. Examples include:

    - Changeout Schedule
    - Changeout Analysis
    - Changeout Analysis – Details
    - Component Status (Suppression Impact)

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

4. Monitor Existing Tasks

Overview

The planner should continue to review and monitor the condition of the components as the strategy due date or usage approaches, to refine the long-term plan. It's helpful to set a reminder in your calendar software to ensure the components are regularly reviewed and monitored.

Another factor affecting strategy tasks is the condition monitoring results (if the CBM module is implemented and configured). The planner should review the accuracy of the estimated due dates/usages for existing tasks.

To review the accuracy of the estimates for existing tasks, group all tasks with the **System Calculated** flag switched off. Condition monitoring alerts display in columns R1 through to R6; each column represents one of the last 6 readings (R) related to the strategy task system or component.

Procedure – Monitor Existing Tasks

Monitor the existing strategy tasks for the accuracy of next occurrences. RVIC and PCV are analysed in this step.

If needed, refine the due date or usage of the estimates by clicking on the task and adjusting the **Due Date/Usage**. You can also move the task along the Gantt chart using the methods discussed in the Review New Tasks section.

Perform the following steps to monitor existing tasks in the Long-term Planning screen:

1. Search **Long-term Planning** using the **Search** field and double-click **Long-term Planning**.



*Alternatively, from the **My Functions** hierarchy, select **Long-term Planning** > **Long-term Planning**.*

2. Remove the **Not Reviewed** status from the **Task Review Status** field.
    i. Set the **Overdue** field as **No**.
    ii. Set **Display Fields** to **Life used %**, **Days Remaining**, **Cost**, **Risk Rating (Calculated)** or **Risk Override**.
    iii. Click **Apply Filter**.



*If you make a change to the strategy date or strategy usage, change the **Review Status** field to **Reviewed** to identify such changes easily.*

If you make a manual change to a strategy task's next occurrence date/usage, AMT prompts you to enter a reason.

*If you want to change the risk rating for the strategy task, select the appropriate risk rating in the **Risk Override** cell after selecting **Risk Override** in **Display Fields**.*

3. AMT displays the corresponding data set in the **Grid** tab. To view the Gantt chart for the same data set, select the **Gantt Chart** tab. Note the following in the Gantt Chart view:

    - The orange boxes represent overdue tasks.
    - The shaded boxes represent future occurrences of the strategy task:
        - Green shaded boxes represent future occurrences driven by suppression.
        - Black shaded boxes represent future occurrences **not** driven by suppression.
    - The red line represents the present day.
    - Purple boxes represent confirmed next occurrences.
    - Dark blue boxes represent next occurrences manually updated directly in AMT.
    - Light blue boxes represent next occurrences forecast by AMT that have not been updated manually in AMT.
    - The timescale can be used to view a date range of up to 2 years on the Gantt chart.

Colour	Description
	Overdue (next occurrence)
	System-calculated and not overdue (next occurrence)
	Not system-calculated, not overdue, not confirmed (next occurrence)
	Confirmed (next occurrence)
	Future occurrences (beyond the next)
	Future occurrence of a strategy task that belongs either to a dependency or suppression scheduling group

5. Set Date for Tasks within Twelve Weeks (3 month look ahead)

Overview

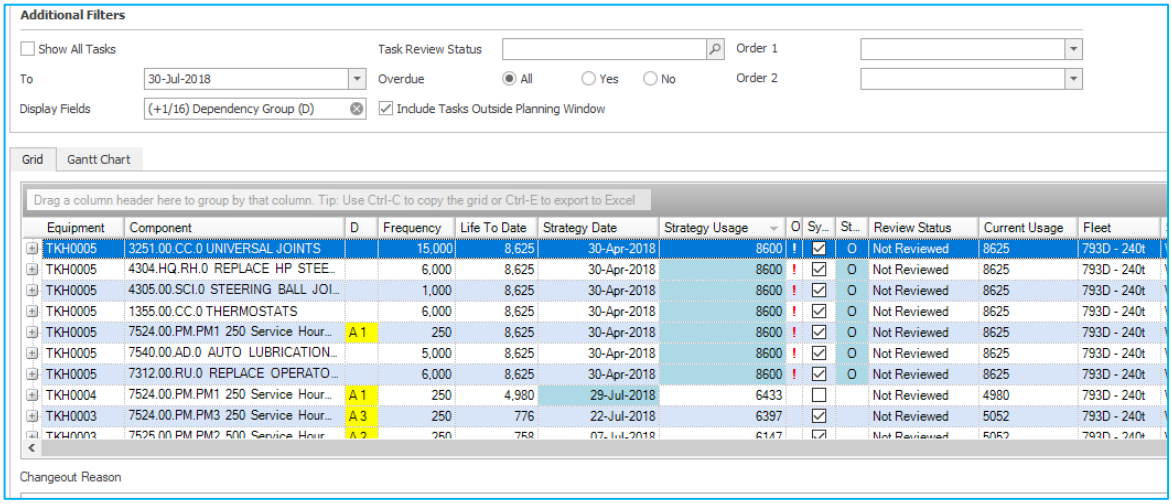
The planner narrows focus on the tasks due in the next 12 weeks. Set the date range on long-term planning to focus on the tasks within this period.

Procedure – Set Date for Tasks

Perform the following steps to set the date for tasks:

1. In the **Long-term Planning** filter screen:
   - Set the **Show all Tasks to Date** field to be 12 weeks from the current date.
   - Click **Apply Filters** to retrieve the relevant records.

   Compare the **Strategy Date** and **Strategy Usage** fields and adjust the values if necessary, using the methods detailed in the previous sections.



Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

6. Optimise Timing (Group/Align Tasks)

Overview

On the Long-term Planning **Gantt** chart, you can visualise tasks and group or align them, improving the scheduling and the equipment's stoppage.

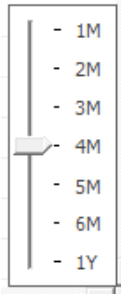
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Procedure – Group Tasks

Perform the following steps to group or align planned strategy tasks in the **Gantt View** of the Long-term Planning screen:

1. Set the zoom to no more than 5 months (**5M**).



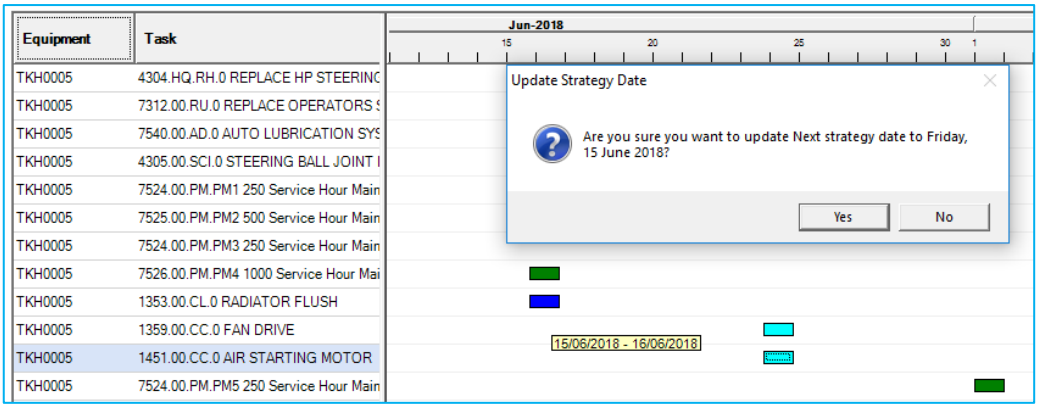
2. Drag the bar along the Gantt chart. To do this, click once on the bar to drag, and then move the mouse until the pointer converts to a four-pointed cross. You can then drag the task bar horizontally.



The Gantt chart must be at a zoom between one month (1M) and four months (5M) to be able to click and drag a bar representing a task.

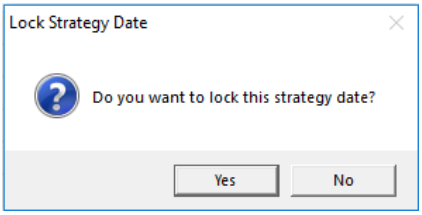
3. The system displays a message asking if you want to update the next strategy date of the task.

   - Click **No** to return the task to its original date.
   - Click **Yes** to display the **Lock Strategy Date** message.

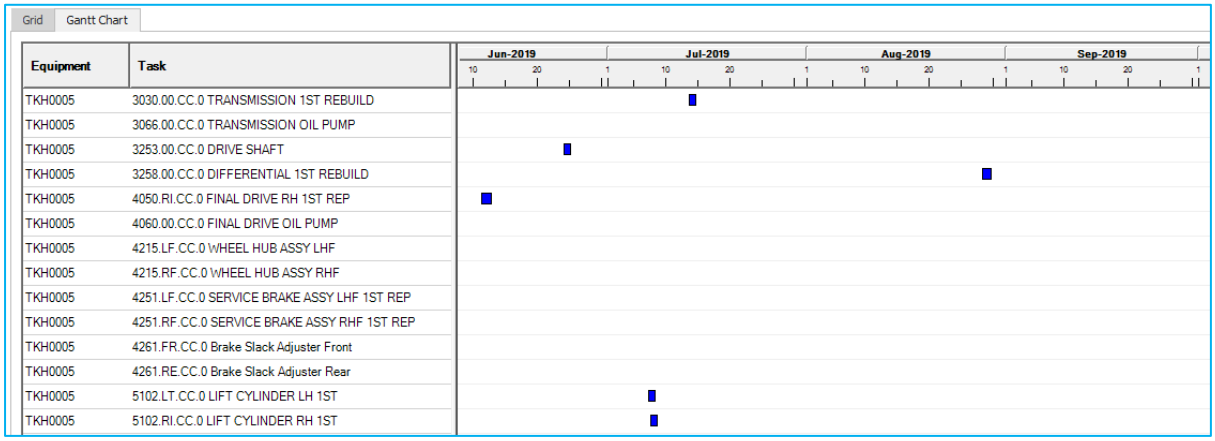


4. The lock in the strategy date notification is displayed:

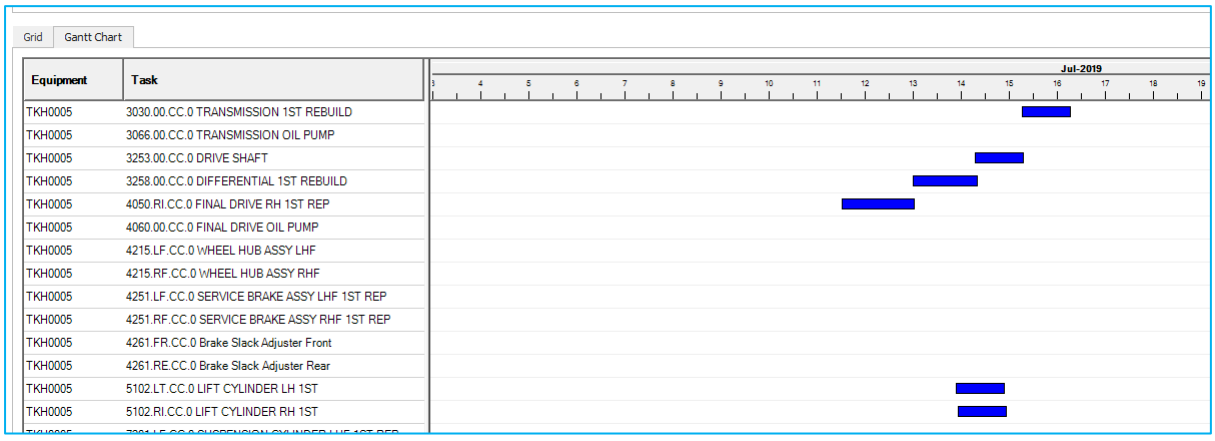
   Click **Yes** to lock in the strategy date, which will also set the **Confirmed** flag against the strategy task's work order in the Planning module. Otherwise, click **No.**



BEFORE CONSOLIDATING (ZOOM 5M)

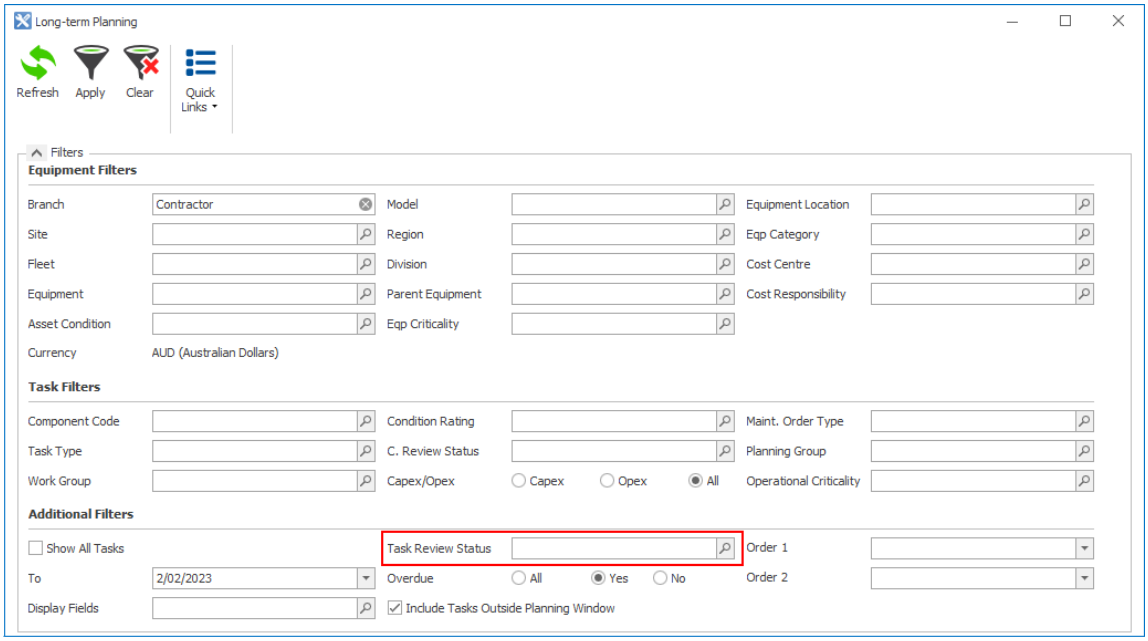


AFTER CONSOLIDATING (ZOOM 1M)

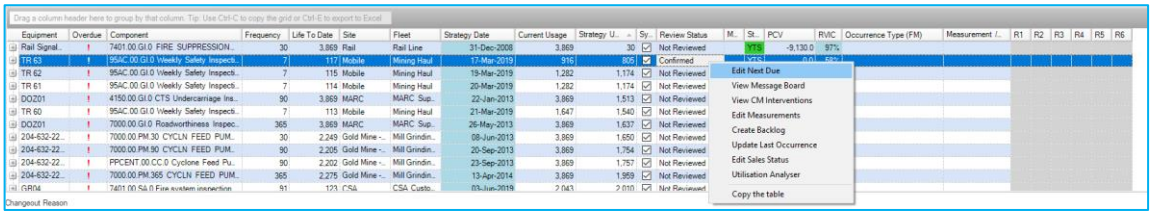


If the next occurrence is accidentally confirmed, perform the following steps to unconfirm it:

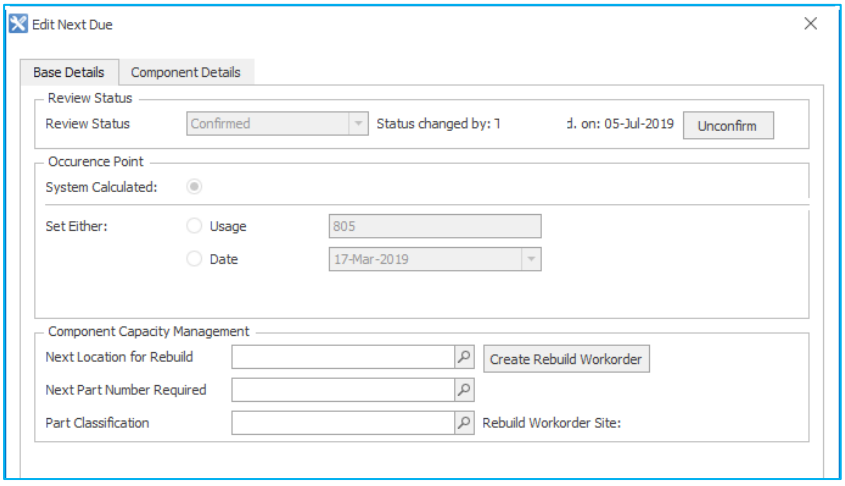
- Filter using the **Task Review Status** = **Confirmed**.



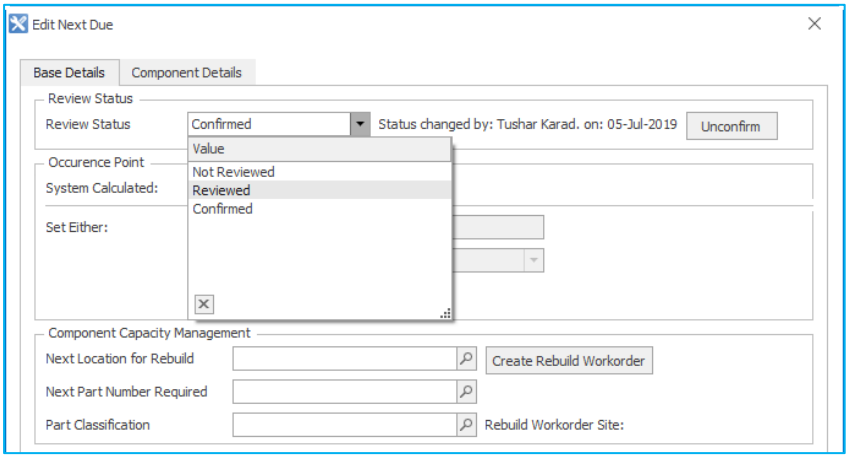
8. Highlight the task to be unconfirmed, then right-click and select **Edit Next Due**.



9. Click **Unconfirm**. You can only do this if you have the necessary permission to unconfirm tasks (contact your AMT Administrator to check).



10. Select the new status, **Reviewed** or **Not Reviewed** and click **Save**.



5. After tasks are consolidated for optimisation purposes, you must evaluate whether the resulting demand placed on resources is realistic. You need to validate:

    - Labour
    - Parts/Components
    - Availability

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

7. Validate Parts and Components

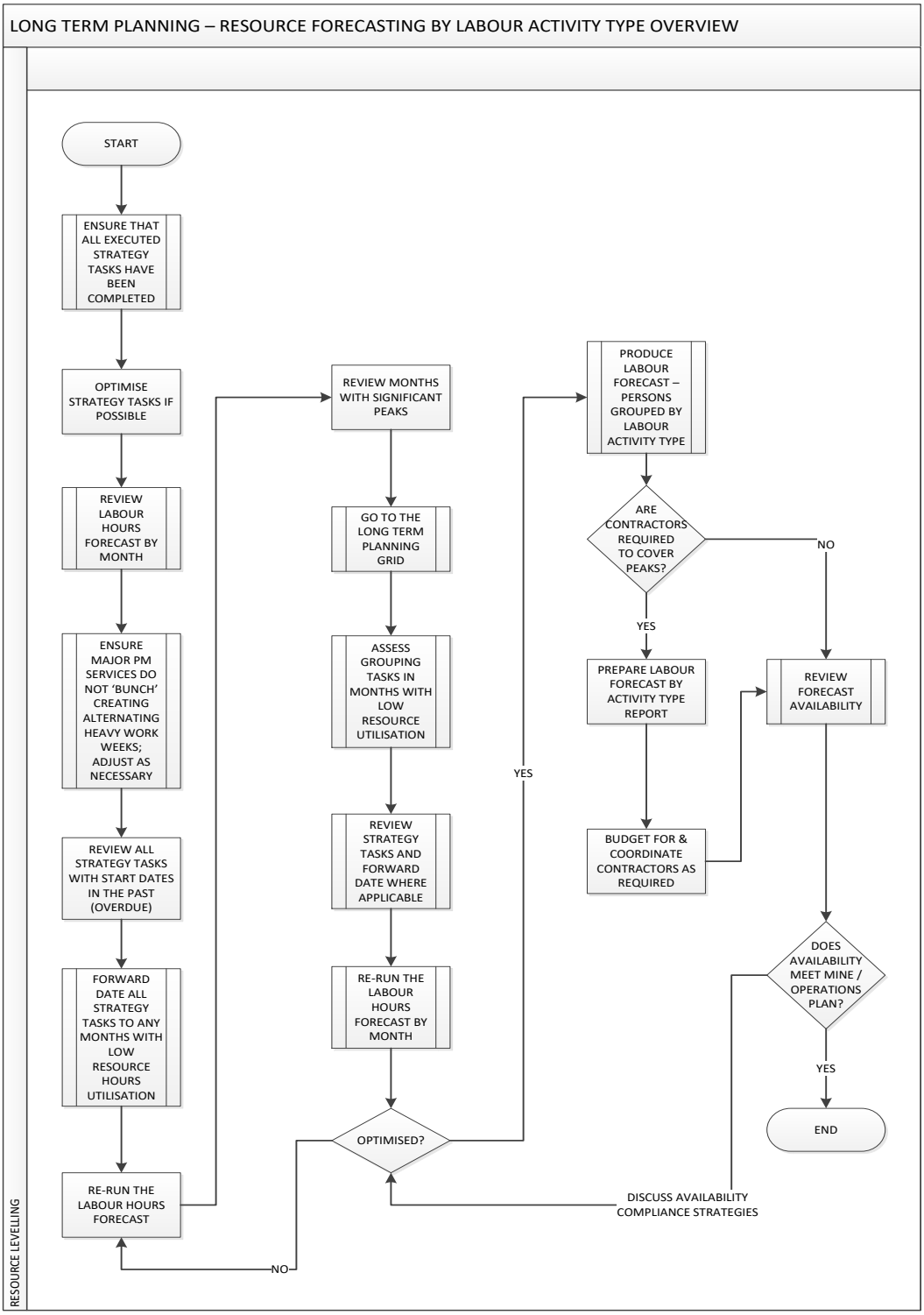
Procedure - Validate Parts and Components

Perform the following steps to run the **Changeout Analysis** report:

1. Open **Long-term Planning**.
2. Under **Components**, locate **Changeout Analysis** reports.
3. Run reports to validate forecasted parts and components.

8. Validate Labour Resources

Process Diagram



Process Inputs

The following process inputs are required to forecast labour effectively:

- Backlog managed correctly (no unexecuted strategy tasks with start dates in the past).
- Standard jobs with labour activity types and associated resource hours by operation.
- Strategy tasks optimised (duration).
- PM services levelled out (smoothing has occurred so that equal amounts of large PM services occur each week; there is no bunching on alternate weeks).
- Where possible, all overdue strategy tasks have been reviewed and rescheduled (future dated) based on the condition of equipment.

Procedure – Validate Labour Resources

Perform the following steps to validate Labour resources:

1. Go to **My Functions** > **Long-term Planning** > **Labour**.
2. Run **Labour Forecast - Hours** to a show bar chart of labour hour requirements.

   Run **Labour Forecast** - **Persons** to show labour skills requirements.
3. Identify tasks and move tasks to balance labour and skill requirements, and rerun reports to review changes.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

9. Register a Manual Component Changeout

Procedure – Register a Manual Component Changeout

Perform the following steps to manually register a component changeout record:

1. Navigate to **Daily Operations** > **Work Order Cost Analysis** > **Work Order Settlement Manager**.
2. Click **Add** to create an AMT internal work order settlement:
3. Select how to generate the work order settlement number. It can be either manually entered or automatically generated by AMT.
4. Search for the relevant equipment using search filters.

   Note the following points:

   - The work order settlement number must be unique if it is entered manually.
   - All work order settlements generated using this process are prefixed with AMT.
   - Availability statistics are calculated using the availability type set at the fleet level.
5. Set the job date or usage when the component changeout happened.

   Set **Job Status** = **Completed** to inform AMT that the component has been changed.
6. Set **Select from Strategy Task = Yes**

   Search for the appropriate strategy task that reflects the component replaced. Click **OK**.

   If known/relevant, enter additional changeout details:

   - Occurrence Type
   - Job Code
   - STO
7. Complete the relevant cost codes and click **Save** when you're finished.



Check with your AMT system administrator or the group responsible for the equipment's maintenance strategy to determine the appropriate cost codes (that is, cost bearer, cost centre and expense element).

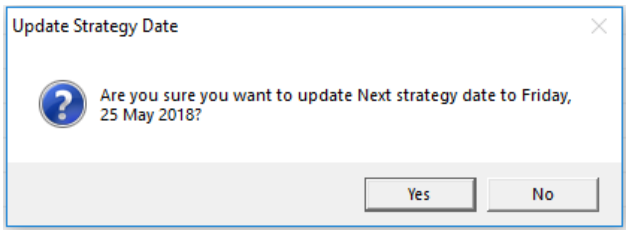
*If parts, labour or miscellaneous costs are going to be entered into the work order settlement, and if the company is not an OEM or dealer, ensure that **Sell** and **Cost** figures are identical*

10. Confirm Component Changeout Schedule

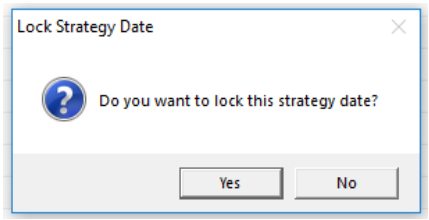
Procedure – Confirm Component Changeout Schedule

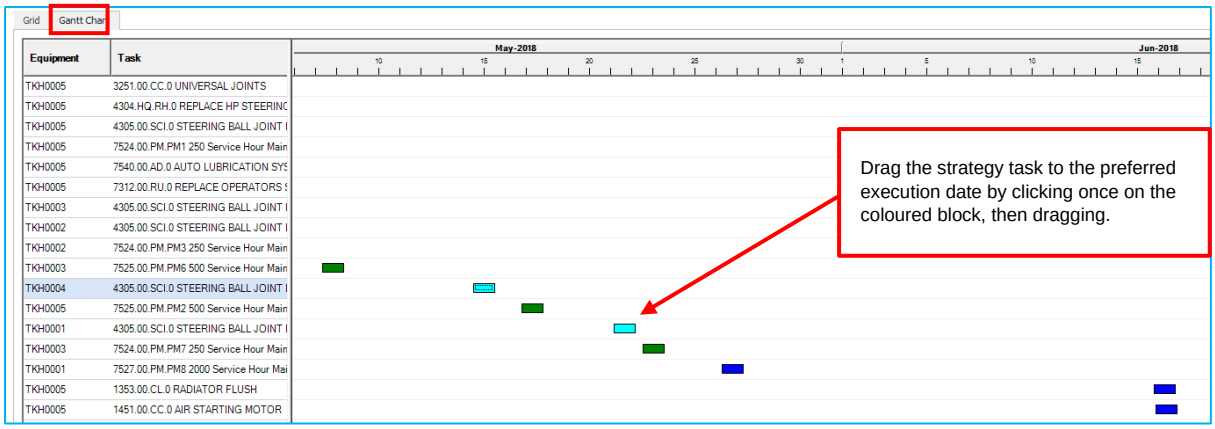
To update and generate a component forecast from AMT, perform the following steps.

1. Navigate to **Long-term Planning**.
2. Filter for the required fleet.
3. Set the following filters to see the strategy tasks requiring immediate attention:
    i. Clear the **Include Tasks outside Planning Window** checkbox.
    ii. Click **Apply Filters**.
4. Note the following points:
    - **Excluding Tasks Outside the Planning Window** means that strategy tasks that have not yet reached their respective lead days are not shown.
    - After a task achieves its lead days, it becomes a priority.
5. To set the next occurrence based on usage (such as engine hours), perform these steps:
    i. Highlight the task in the grid.
    ii. Click **Strategy Usage**.
    iii. Enter the target usage (the equipment's usage when the strategy task will be performed).
6. Alternatively, to set the next occurrence based on date, perform these steps:
    i. Highlight the task in the grid.
    ii. Click **Strategy Date**.
    iii. Enter the target date when the strategy task will be performed.
7. Alternatively, to set the target date:
    i. Click the **Gantt Chart**.
    ii. Click the strategy task in the task list.
    iii. Click the strategy task box. When the cursor changes to a 4 headed arrow  :
        a. Slide the strategy task along the axis to the target date.
    iv. To update the next strategy date, click **Yes**.



    v. To lock in the strategy date, click **Yes**.





8. After a new execution date has been assigned, change the Review Status to be either:

   - **Reviewed** - typically a target changeout point has been entered into the system, but, for example, the component's health is still being assessed, so the final decision has not yet been made.
   - **Confirmed** - the target date or usage is accurate and planners should work to this target.

   A sample of reports that will show the adjustments made in long-term planning (**Long Term Planning** > **Components>)** are:

   - Change out Schedule
   - Change out Analysis
   - Change out Analysis – Details
   - Component Status

   Some of these reports are also available by clicking the **Quick Links** menu.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

11. Review Forecasts

Overview

In the last step, you review forecasts to identify risk and performance impacts.

AMT has a set of forecast reports to use to review the long-term plan. These reports drive not only long-term planning but also the full life cycle forecast of the equipment.

The reports are:

- Forecast Availability
- Cost forecast reports:
    - Forecast Cost Analysis
    - Forecast Worksheet
    - Cost Analysis - “Life Cycle”
- Component forecast reports:
    - Component Status
    - Overdue Strategy Tasks
    - Changeout Analysis - Detailed
- Labour forecast reports
    - Labour Forecast - Hours
    - Workload Optimisation

12. Forecast Availability Report

Overview

If the asset is based on usage, and the fleet to which the asset belongs is enabled for availability forecasting, you can run the Forecast Availability report.

Procedure – Forecast Availability Report

Perform the following steps to review forecasts using the Forecast Availability report:

1. Navigate to **Fleet Administration** and click **Filter**.



*Alternatively, from the **My Functions** hierarchy, select **Setup & Admin** > **Equipment** > **Fleet Administration.***

2. Highlight the fleet and double-click it.
3. Click **Availability Calculation**.



*Availability statistics are calculated using the availability type set at the fleet level. Valid availability types are **Physical** or **Mechanical**. Owner maintainers typically select physical availability, while equipment hire companies and maintenance and repair contracts typically select mechanical.*

4. To enable the fields, select **Enable Forecasting Availability**.
5. When the fields are active, set the following parameters:
    - **Annual Schedule Hours - Assumption** - how many hours each equipment in the fleet will operate per annum.
    - **Scheduling Efficiency % - Assumption** - how efficient maintenance planning will be performed. That is, will the planner group all major tasks that are forecast to occur in the same month, to be executed at the same time (the absolute minimum stoppages ~ 100%) or will each task be performed at an independent stoppage (maximum possible stoppages ~ 0%)?
    - **Task Specific Downtime** - the task types that should be factored into the availability forecast. For example, if 95% is set as the annual base downtime rate, and you select component changeouts and PM services to be included (because they will be performed), the availability rate will drop. The downtime for changeouts and PM services will be factored into the availability forecast.
    - **Annual Base Downtime** - the baseline availability expected per year if only minor breakdown-type repairs were to be performed. Usually the rate is high, for example 95%.
6. Click **Save**.



If availability has been optimised to the point that labour resources levels are being increased, you need to examine the situation with Maintenance management and develop short-term tactics to help overcome the shortfalls. For example, hiring in additional equipment temporarily, or bringing in additional labour for key periods identified, or a combination of both.

7. Search **Forecast Availability** using the **Search** field and double-click **Forecast Availability**.



*Alternatively, from the **My Functions** hierarchy, select **Long-term Planning > Availability > Forecast Availability.***

8. On the **Forecast Availability - Report Parameters** dialog, select the relevant equipment and click **Preview**.

9. The **Forecast Availability** report displays.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

13. Cost Forecast Reports

Overview

When running cost reports, be mindful of selecting the cost bearer as maintenance and what should be included or excluded in terms of parts, labour, or miscellaneous costs.

Procedure – Forecast Cost Analysis report

The Forecast Cost Analysis report shows in a graphical view the strategy task costs at sell.

Perform the following steps to review forecasts using the Forecast Cost Analysis report:

1. Search **Forecast Cost Analysis** using the **Search** field and double-click **Forecast Cost Analysis.**



*Alternatively, from the **My functions** hierarchy, **select Long-term Planning** > **Costs** > **Forecast Cost Analysis.***

2. In the **Analyse By** field, select **Equipment**.
   i. Select the relevant **Fleet**.
   ii. Click **Preview**.



- *We recommend that you manually set the date range for the report from the current date to the next 12 months. Before manually setting the date range, ensure the **Entire Plan Term** checkbox is clear.*
- *In the **Date Selector** section, the **Entire Plan Term** checkbox refers to All History + Future and the **Plan To Current Month** checkbox refers to the Cost to Date.*
- *Users can set the desired date range depending on their requirements. There is no prescription.*

3. The **Cost Analysis** report displays. This report shows both costs to date and the future costs to the end of the asset's life.

Procedure – Forecast Cost Worksheet report

The Forecast Worksheet report is used for forecasting costs by expense group. It shows future and historical costs at sell, categorised by up to five different areas.

Perform the following steps to review forecasts using the Forecast Worksheet report:

1. Search **Forecast Worksheet** using the **Search** field and double-click **Forecast Worksheet**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning > Costs** > **Forecast Worksheet**.*

2. On the **Forecast Worksheet - Report Parameters** dialog, perform the following actions:
   i. Select the **Next 12 Months** checkbox.
   ii. For the **Group By** field, select **Site**, **Cost Centre**, **Parts/Labour/Misc** and **Expense Element**.
   iii. Select the relevant **Fleet**.
   iv. Click **Preview**.

   The Forecast Worksheet report displays. This report shows both costs to date and the future costs to the end of the asset's life.

Procedure – Cost Analysis – "Life Cycle" report

The Cost Analysis - "Life Cycle" report shows a high-level cost per month for each asset. It is run the same way as the Forecast Cost Analysis report, but you can drill down in this report.

Perform the following steps to review forecasts using the Cost Analysis - "Life Cycle" report:

1. Search **Cost Analysis - "Life Cycle**" using the **Search** field and double-click **Cost Analysis - "Life Cycle"**.



*Alternatively, from the **My Functions** hierarchy, select **Strategy Optimisation** > **Cost Analyses** > **Cost Analysis - "Life Cycle".***

2. On the **Cost Analysis - "Life Cycle" - Report Parameters** dialog, perform the following actions:
   i. In the **Analyse By** field, select **Equipment**.
   ii. Select the relevant equipment.
   iii. Set the **Projection Type** as **Current** and **Cost Bearer** as **Maintenance**.
   iv. Select the **To Date** and **Future** checkboxes in the **Costs** field.
   v. Select the **Parts**, **Labour** and **Misc** checkboxes.
   vi. Set the relevant **Currency**.
   vii. Click **Preview**.

   The Cost Analysis - "Life Cycle" report displays.

3. In the bar across the top of the screen, select **Strategy Task** from the list.
4. Click either one of the wedges of the pie chart or one of the rows in the table. This will trigger the drill down.

   The corresponding Cost Analysis - “Life Cycle” report displays.
5. Click any of the **Workorders** hyperlinks to view associated work orders.



Click the blue back arrow  *to go back one step.*

The Forecast Worksheet report displays. The costs shown will be forecast.

14. Component Forecast Reports

Procedure – Component Status report

The Component Status report shows a component overview of the machine, including overdue components, next changeouts and historical workorders. This report is run for a selected equipment.

Perform the following steps to review forecasts using the Component Status report:

1. Search **Component Status** using the **Search** field and double-click **Component Status**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning** > **Components** > **Component Status.***

2. On the **Component Status – Report Parameters** dialog:
    i. In the **Analyse By** field, select **Equipment**.
    ii. Select the relevant equipment and fleet.
    iii. Click **Preview**.
3. The Component Status report displays. This report shows the current life of each strategy, when the next changeout is due and when the last changeout occurred.

Procedure – Overdue Strategy Tasks report

The Overdue Strategy Tasks report displays the overdue components, and the costs associated with overdue tasks. These components need to be rescheduled and a decision made as to when these are expected to be done.

Perform the following steps to review forecasts using the Overdue Strategy Tasks report:

1. Search **Component Status** using the **Search** field and double-click **Overdue Strategy Tasks**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning** > **Components** > **Overdue Strategy Tasks.***

2. On the **Overdue Strategy Task - Report Parameters** dialog, perform the following actions:
    i. Select the relevant **Fleet** to run the report at the fleet-level.
    ii. Click **Preview**.

    The Overdue Strategy Task report displays. The report shows all strategy tasks in the past that are still to be rescheduled and completed.

Procedure – Changeout Analysis - Detailed report

The Changeout Analysis - Detailed report provides a graphical view of the number of components to be changed within a required period. The report provides five levels of Analyse By.

Perform the following steps to review forecasts using the Changeout Analysis - Detailed report:

1. Search **Changeout Analysis - Detailed** using the **Search** field and double-click **Changeout Analysis - Detailed**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning** > **Components** > **Changeout Analysis - Detailed.***

2. On the **Changeout Analysis - Detailed - Report Parameters** dialog, perform the following actions:
    i. Select **Entire Plan Term**.
    ii. Select the relevant **Fleet**.
    iii. In the **Analyse By** field, select **Site** and **Strategy Task**.
    iv. Select the relevant **Task Type**.
    v. Apply any other relevant filters.
    vi. Click **Preview**.

    The Changeout Analysis - Detailed report displays.

15. Labour Forecast Reports

Procedure – Labour Forecast - Hours report

Use the Labour Forecast - Hours report to analyse maintenance work by hours required. Group the report to a maximum of five categories for more specific analysis.

Perform the following steps to review forecasts using the Labour Forecast - Hours report:

1. Search **Labour Forecast - Hours** using the **Search** field and double-click **Labour Forecast – Hours**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning** > **Labour** > **Labour Forecast - Hours***

2. On the Labour **Forecast - Hours - Report Parameters** dialog, perform the following actions:
    i. Set the date range as the next 12 months.
    ii. In the **Group By** field, select **Labour Activity**.
    iii. Select the relevant **Site**.
    iv. Click **Preview**.

    The Labour Hours Forecast report displays.

Procedure – Workload Optimisation report

Use the Workload Optimisation report to compare labour hours by month, quarter or year for two or more projections.

Perform the following steps to review forecasts using the Workload Optimisation report:

1. Search **Workload Optimisation** using the **Search** field and double-click **Workload Optimisation**.



*Alternatively, from the **My functions** hierarchy, select **Long-term Planning** > **Labour** > **Workload optimisation.***

2. On the Workload Optimisation - Report Parameters dialog, perform the following actions:
    i. Set the date range.
    ii. Set **Analysed By** to **Month**.
    iii. Set **Year Start Month** to **January**.
    iv. In the **Group By** field select **Equipment Plan**.
    v. Select the relevant **Fleet**.
    vi. Select **Projection Type** to **Current**.

    The Workload Optimisation report displays.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

16. Long Term Planning Meeting

Functionality Overview

After you're satisfied that the labour resources will be available to support the plan, and that the mine plan availability KPIs are met, management should approve the plan.

ATTENDEES

Suggested meeting attendees:

- Production manager
- Maintenance manager
- Reliability engineer
- Workshop manager
- Parts or logistics representative (including rebuild centre if applicable)

REPORTS

Use the following reports from AMT:

- Component Forecast
  - **Changeout Analysis – Detailed**
  - **Technical Performance Analysis**
- Labour Forecast
  - **Labour Forecast – Hours** (by Labour Activity Type); or
  - **Labour Forecast – Persons** (by Labour Activity Type)
- Availability Forecast
  - **Forecast Availability**
  - **Asset Utilisation** (forecast)

17. Confirm Plan

Functionality Overview

After the long-term planning meetings and weekly maintenance scheduling meetings, the Long-term Planner may need to update thelong-term plan's parts, labour and availability validations.

Where applicable, follow sections 1 through to 10 to adjust the long-term plan.

18. Communicate to Stakeholders

Functionality Overview

Now that the plan is confirmed, you should email the following documents to the following people.

Document	Source	Distribution List
Long-term Plan	AMT > Changeout Schedule AMT > Changeout Analysis AMT > Changeout Analysis - Detailed	Maintenance Manager Reliability Engineer Planner Logistics/Purchasing Officer
Labour Requirements	AMT > Labour Forecast – Hours AMT > Labour Forecast - Persons	Maintenance Manager Maintenance Superintendent Supervisor Planner
Availability Forecast	AMT > Forecast Availability	Production/Operations Manager Maintenance Manager Production/Operations Superintendent Maintenance Superintendent Planner

19. Summary

Summary

The key points to remember from this guide are:

- From an AMT perspective, the long-term planning process involves reviewing and optimising the next occurrences of planning strategy tasks, as they normally have a significant impact on the predicted maintenance cost spend in a budget or forecast.
- You can perform the following long-term planning steps in AMT:
    - Review actual usage readings
    - Review overdue tasks
    - Review new tasks
    - Monitor existing tasks
    - Optimise Timing (Group/Align Tasks)
    - Validate Parts and Components
    - Validate Labour Resources
    - Confirm Component Changeouts
    - Review forecasts
    - Run Forecast Availability Report
    - Run Cost Forecast reports
    - Run Component Forecast reports
    - Run Labour Forecast reports
    - Long-term Planning meeting
    - Confirm Plan
    - Communicate to Stakeholders